Assignment 11

DNS Server Configuration

1. **Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.**

Requirements:

- Windows 11
- Simple DNS Plus software
- Internet connection

Procedure:

Step 1: Download and Install

1. I opened Google Chrome and searched for Simple DNS Plus.
2. I visited its official website and downloaded the installer.
3. I opened the setup file and completed the installation.
4. I launched Simple DNS Plus.

Step 2: Create a DNS Zone

1. **I opened the Records section.**
2. **I selected the New Zone option.**

3. **I selected Primary Forward Zone.**
4. **I entered the domain name musicmania.in.**
5. **I clicked Finish to create the new zone.**

Step 3: Verify the Zone

1. **I checked the zone list in Simple DNS Plus.**
2. **I confirmed that musicmania.in was successfully created.**
3. **The DNS server was ready for adding DNS records.**

Result:

Successfully installed and configured Simple DNS Plus on Windows and created a new Primary Forward DNS Zone named musicmania.in.

2. **Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.**

Step 1: Add an A Record

1. I opened the Simple DNS Plus application.
2. I selected the musicmania.in DNS zone.
3. I clicked New Record and selected A (Host) Record.
4. I entered the hostname www.
5. I entered the IP address 192.168.1.50.
6. I saved the record.

Step 2: Verify Using nslookup

1. I opened Command Prompt (CMD).
2. I entered the following command:

---

nslookup www.musicmania.in

---

3. I checked the output to verify that the domain resolved to 192.168.1.50.

Expected Output:

Name: www.musicmania.in

Address: 192.168.1.50

Result: Successfully created an A record for www.musicmania.in pointing to 192.168.1.50 and verified the DNS mapping using the nslookup command.

3. **Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.**

Procedure:

Step 1: Add a CNAME Record

1. I opened the Simple DNS Plus application.
2. I selected the musicmania.in DNS zone.
3. I clicked New Record and selected CNAME Record.
4. I entered play as the hostname.
5. I entered www.musicmania.in as the target hostname.
6. I saved the record.

Step 2: Verify Using nslookup

1. I opened Command Prompt (CMD).
2. I entered the following command:

```
nslookup play.musicmania.in
```

3. I checked whether the domain resolved to the IP address 192.168.1.50.

Expected Output:

```
Name:    www.musicmania.in

Address: 192.168.1.50
```

Result: Successfully created a CNAME record for play.musicmania.in, pointing to www.musicmania.in. The DNS lookup should resolve to 192.168.1.50.

4. **Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig. Hint: Include at least three different troubleshooting commands or checks in your documentation.**

Procedure:

Step 1: Check DNS Configuration

1. I opened Simple DNS Plus.
2. I checked whether the musicmania.in zone was configured correctly.
3. I verified the A record and CNAME record for any mistakes.

Step 2: Check DNS Service

1. I opened Windows Services by typing services.msc in the Run dialog.
2. I located the Simple DNS Plus service.
3. I checked whether the service was running.
4. If stopped, I restarted the service.

Step 3: Check DNS Using Commands

I opened Command Prompt (CMD) and used these commands:

1. Check IP configuration:

---

ipconfig /all

It displays the DNS server address configured on my computer.

2. Clear DNS cache:

---

ipconfig /flushdns

It clears old DNS records stored on my computer.

3. Test domain resolution:

---

nslookup www.musicmania.in

It checks whether the domain resolves to the correct IP address.

4. Test using a specific DNS server:

---

nslookup www.musicmania.in 127.0.0.1

It sends the DNS query directly to the local DNS server.

Step 4: Verify the Result

- I checked whether www.musicmania.in resolved to the correct IP address 192.168.1.50. If it worked, the DNS configuration was functioning correctly.

Result: Successfully identified the possible DNS configuration and service issues and followed troubleshooting steps to restore domain name resolution.